MILITARY SPECIFICATION SHEET

INTERCOMMUNICATIONS SETS, AN/AIC-14 and AN/AIC-14A

ACTIVE INTERPHONE CONTROL PANEL

C-2646/AIC-14

Inactive for new design after 10 August 1984

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the unit described herein shall consist of this document and the latest issue of specification MIL-I-22352(AS).

1. SCOPE

1.1 Scope.

This specification sheet provides performance and physical characteristics of the active Interphone Control panel C-2646/AIC-14, hereinafter referred to as C-2646.

1.2 Military Part Number.

The military part number shall be M22352/19-1

2. APPLICABLE DOCUMENTS

Refer to 20 of MIL-I-22352(AS) Appendix A for a listing of applicable documents. The requirements specified herein shall have precedence over all referenced documents. The contract and its attachments shall have precedence over any specification or referenced document.

3. REQUIREMENTS

3.1 Item definition. The active Interphone Control panel C-2646 shall provide:

(1) Amplification of microphone audio for outgoing interphone signals.
(2) Current supply to power a microphone preamplifier or carbon microphone.

FSC 5831

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(3) Keyswitch and connection of microphone generated interphone signals to the interphone output.
(4) Amplification of Common Audio signals.

3.1.1 Item diagram. The C-2646 active Interphone Control panel shall provide the functions represented schematically in accordance with the diagram of Figure 1.

3.1.2 Interface definition. The C-2646 active Interphone Control panel input and output signals will normally exhibit the characteristics presented in Table I.

TABLE I. Signals characteristics.

Signal Name	Characteristics
Inputs:	
Microphone Audio (MIC AUDIO)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0.01 to 18 Vac.
Auxiliary Interphone (AUX INTPH)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0.01 to 0.1 Vac.
Common Audio (COMMON AUDIO)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 15 Vac.
External Transmit Key (EXT KEY)	Relay logic signal: 1) Open 2) Short to ground
Power (PWR)	+27.5 ±0.5 Vdc
Outputs:	
Transmitter Key (XMIT KEY)	Relay logic signal: 1) Open 2) Short to ground
Interphone (INTPH)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0.01 to 18 Vac.
Headset (HDST)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 18.0 Vac.
Common Audio (COMMON AUDIO)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 1.0 Vac.

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TABLE I. Signals characteristics - continued.

Signal Name	Characteristics
Grounds:	
Power (PWR GND)	Power returns. See 3.4.4.2 of MIL-I-22352(AS).
Panel Lights Return (LIGHTS RTN)	Indicator return to power ground.
System (SYS GND)	Signal return. See 3.4.4.3 of MIL-I-22352(AS).
Microphone Return (MIC RTN)	Signal return to power ground.
Headset Return (HDST RTN)	Signal return to power ground.

3.1.3 SRAs. The C-2646 shall contain an audio amplifier SRA having a gain of 43 ± 3 dB at 1000 Hz. Performance shall be in accordance with MIL-I-22352/1(AS).

3.1.4 Controls. The C-2646 active Interphone Control unit shall provide an Interphone Volume Control (INTPH VOL).

The INTPH VOL control shall be a rotary type control, providing an adjustment range of 30 ± 3 dB. The amplitude of incoming radio receiver signals shall not be affected by more than ± 1 dB when the INTPH VOL control is adjusted throughout its range.

3.2 Characteristics

3.2.1 Performance characteristics. This section specifies the minimum performance requirements for each of the inputs, outputs, and functions of the active Interphone Control panel. Unless otherwise specified, selection of any function or mode of the active Interphone Control panel shall not compromise the performance of any input or output of the IC system.

3.2.1.1 Input characteristics. The inputs listed in Table I shall be provided having the following characteristics:

3.2.1.1.1 Microphone Audio (MIC AUDIO).

3.2.1.1.1.1 Signal Frequency. The MIC AUDIO signal input shall accept input drive signals which range from 300 to 6,000 Hz.

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3.2.1.1.1.2 Signal Amplitude. The MIC AUDIO signal input shall accept input drive signals which range in amplitude from 0.01 and 18 Vac. The circuitry shall be capable of withstanding overvoltage inputs of up to 20 Vac without sustaining damage.

3.2.1.1.1.3 Input Impedance. The impedance of the MIC AUDIO signal input shall be 100 ohms ± 10 percent.

3.2.1.1.1.4 Microphone Current. The MIC AUDIO input line shall be capable of supplying 50 ± 10 milliamperes to the microphone circuit for carbon microphones or as a source of power to equivalent dynamic microphone and associated preamplifier combinations.

3.2.1.1.2 Auxiliary Interphone (AUX INTPH). The AUX INTPH input shall be applied to the audio amplifier of detail specification MIL-I-22352/1(AS) through the INTPH VOL control and the Key Relay (when deenergized). The AUX INTPH shall have an input impedance of 1,200 Ohms ± 10 percent.

3.2.1.1.3 Common Audio (COMMON AUDIO). The COMMON AUDIO may act as an input or as an output. The COMMON AUDIO input is summed with the AUX INTPH input, passed through the INTPH VOL control and applied to the Amplifier Module when the Key Relay is deenergized. See 3.2.1.2.2 for output characteristics. The COMMON AUDIO input shall have the following characteristics:

3.2.1.1.3.1 Signal Frequency. The COMMON AUDIO input shall accept input drive signals which range from 300 to 6,000 Hz.

3.2.1.1.3.2 Signal Amplitude. The COMMON AUDIO signal input shall accept input drive signals which range in amplitude from 0 to 1.0 Vac.

3.2.1.1.3.3 Input Impedance. The impedance of the COMMON AUDIO input shall be 62,000 ohms ± 10 percent.

3.2.1.1.4 External Transmit Key (EXT KEY). The EXT KEY signal is a relay logic drive signal having two states based on the position of the operator's keyswitch: (1) open and (2) short to ground. The signal shall be applied to input circuitry to activate the keyswitch relay function. Signal current shall not exceed 100 ma.

3.2.1.1.5 Power (PWR). The C-2646 shall accept 27.5 ± 0.5 Vdc power. In operation, all signal inputs open circuited, the input current shall not exceed 500 ma.

3.2.1.2 Output characteristics. The outputs listed in Table I shall be provided having the following characteristics:

3.2.1.2.1 Transmitter Key (XMTR KEY). The XMTR KEY output shall be a relay logic signal having two states (1) open and (2) short to ground. The XMTR KEY signal shall be a short to ground when transmitter keying is desired as evidenced by actuation of the microphone keyswitch.

3.2.1.2.2 Common Audio. The COMMON AUDIO output may also be used as an input. See 3.2.1.1.3 for information on input functions.

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As an output, the COMMON AUDIO line shall provide amplified MIC AUDIO signals when the Key Relay is energized by a logic LO on the EXT KEY input. The COMMON AUDIO output shall have the following characteristics:

3.2.1.2.2.1 Signal Frequency. The COMMON AUDIO output signal amplitude shall be in accordance with the response curves of Figure 2 when terminated in a 600 ohm resistive load.

3.2.1.2.2.2 Signal Amplitude/Gain. The COMMON AUDIO output shall exceed a nominal input of 0.06 V at 1,000 Hz applied to MIC AUDIO input by 43 ± 3 dB. The input/output characteristic (gain) of the amplifier shall be linear within ±2 dB when the input level is varied between 0.01 and 0.10 volts rms.

3.2.1.2.2.3 Distortion. The total harmonic distortion over the frequency range of 300 to 6,000 Hz and input amplitude range of 0.01 to 0.10V shall not exceed 5 percent under standard conditions as defined in MIL-T-5422. The total harmonic distortion over the frequency range of 300 to 6,000 Hz shall not exceed 8 percent throughout the environmental envelope of MIL-E-5400 for Class 2 equipment when operated at rated power.

3.2.1.2.2.4 Output Power. The output of the amplifier shall be capable of delivering 300 milliwatts into a 600 ohm resistive load.

3.2.1.2.2.5 Output Stability. The output shall show no evidence of regeneration or oscillation throughout the environmental envelope of MIL-E-5400 for Class 2 equipment under any combination of input-output open, short, or complex impedance terminations. The C-2646 shall be capable of withstanding any combination of open/short circuit terminations without damage and resume operation within specification limits when properly terminated.

3.2.1.2.2.6 Internal Noise. The amplifier shall have an output of less than 5 millivolts when operated with the input shorted to signal ground throughout the environmental envelope of MIL-E-5400 for Class 2 equipment.

3.2.1.2.2.7 Susceptibility to Stray Magnetic Fields. The amplifier module shall be shielded or so designed that when oriented in any position within a magnetic field of 2 Gauss in the frequency range of 400 to 2,800 Hz, the output from each amplifier circuit when properly terminated shall not be greater than 10.0 millivolts at 400 Hz. Between 400 and 2,800 Hz the output shall not exceed a linear increase up to 50 millivolts at 2,800 Hz.

3.2.1.2.2.8 Output Impedance. The amplifier shall be capable of delivering the specified output power when terminated into a 600 ohm resistive load.

3.2.1.2.3 Headset (HDST).

3.2.1.2.3.1 Signal Frequency. The HDST output shall be level within ±3 dB of the reference level at 1,000 Hz within the frequency range from 300 to 6,000 Hz.

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3.2.1.2.3.2 Signal Amplitude/Gain. The output shall exceed a nominal input at the SEL RCVR input of 0.06 V at 1,000 Hz by 43 ± 3 dB when terminated in a 600 ohm resistive load and the Key Relay is deenergized.

3.2.1.2.3.3 Output Impedance. The impedance of the HDST output shall be 600 ohms ± 10 percent.

3.2.1.2.3.4 Distortion. The total harmonic distortion over the frequency range of 300 to 6,000 Hz shall not exceed 5 percent under standard conditions as defined in MIL-T-5422. The total harmonic distortion over the frequency range of 300 to 6,000 Hz shall not exceed 10 percent throughout the environmental envelope of MIL-E-5400 for Class 2 equipment.

3.2.1.2.3.5 Output Power. The output of the amplifier shall be capable of delivering 300 milliwatts into a 600 ohm resistive load.

3.2.1.2.3.6 Output Stability. The output shall show no evidence of regeneration or oscillation throughout the environmental envelope specified herein under any combination of input-output open, short, or complex impedance terminations. The C-2646 shall be capable of withstanding any combination of open/short circuit terminations without damage and resume operation within specification limits when properly terminated.

3.2.1.2.3.7 Susceptibility to Stray Magnetic Fields. The amplifier module shall be shielded or so designed that when oriented in any position within a magnetic field of 2 Gauss in the frequency range of 400 to 2,800 Hz, the output from each amplifier circuit when properly terminated shall not be greater than 10.0 millivolts at 400 Hz. Between 400 and 2,800 Hz the output shall not exceed a linear increase up to 50 millivolts at 2,800 Hz.

3.2.1.2.3.8 Output Impedance. The amplifier shall be capable of delivering the specified output power when terminated in a 600 ohm resistive load.

3.2.2 Physical characteristics.

3.2.2.1 Dimensions. The size of the unit is shown in Figure 3. All output dimensions shall be within ±0.03 inches of the nominal value shown.

3.2.2.2 Weight. The C-2646 shall weigh no more than 1.5 pounds.

3.3 DESIGN AND CONSTRUCTION

The design and construction of the C-2646 active Interphone Control panel shall be in accordance with the requirements of 3.3 of MIL-I-22352(AS).

4. QUALITY ASSURANCE PROVISIONS

See MIL-I-22352(AS).

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4.1 Responsibility for tests and inspections. See MIL-I-22352(AS).

4.2 Test plan. A detailed test plan will be submitted by the contractor for approval by the acquiring activity which details the tests to be performed in order to verify compliance with the performance characteristics specified in 3.2.1. The test plan will consider all of the inspections delineated in 4.2 of MIL-I-22352(AS).

4.3 Test procedures. Detailed test procedures will be submitted by the contractor for approval by the acquiring activity which will satisfy the requirements of the test plan of 4.2.

Preparing Activity:
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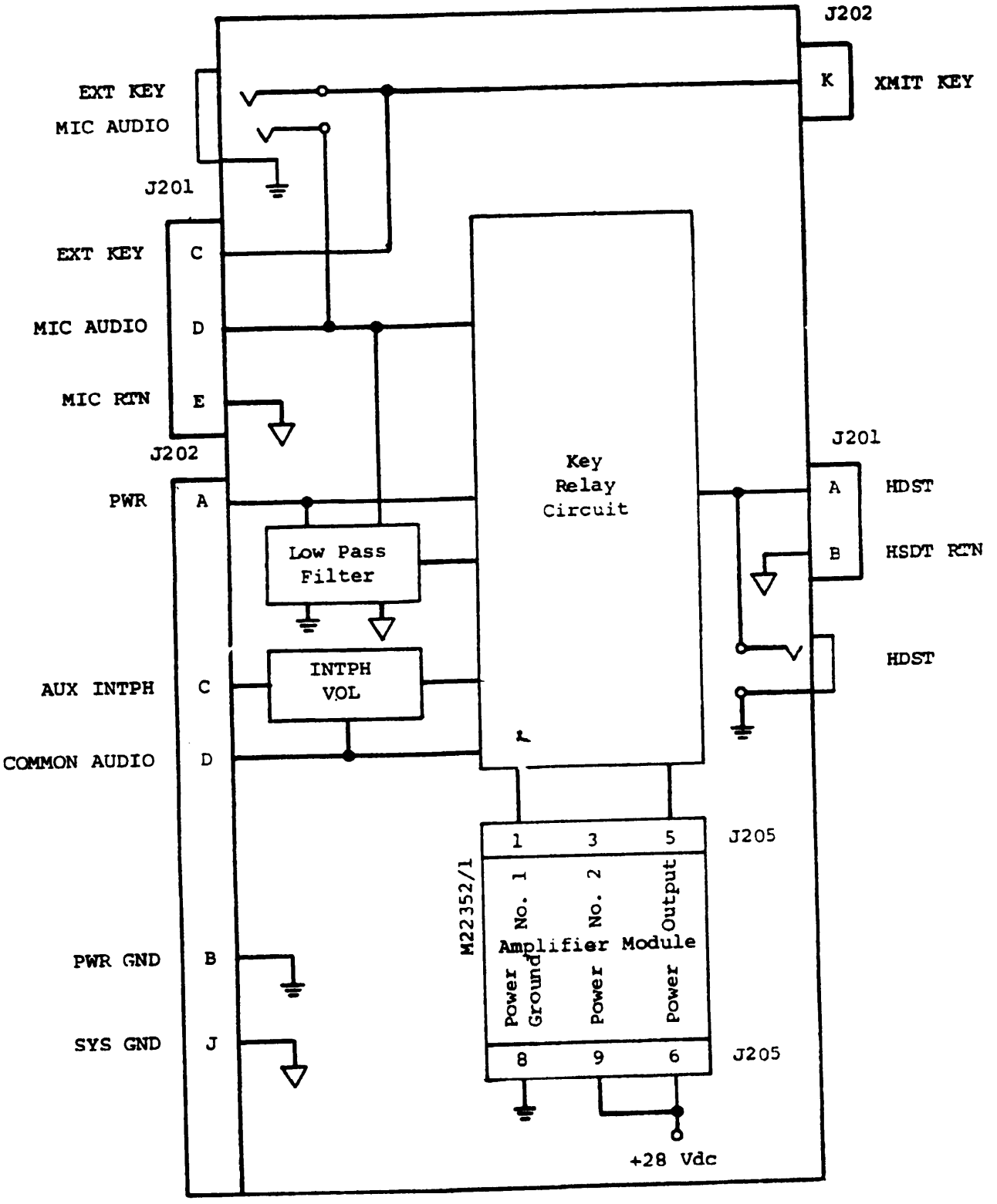
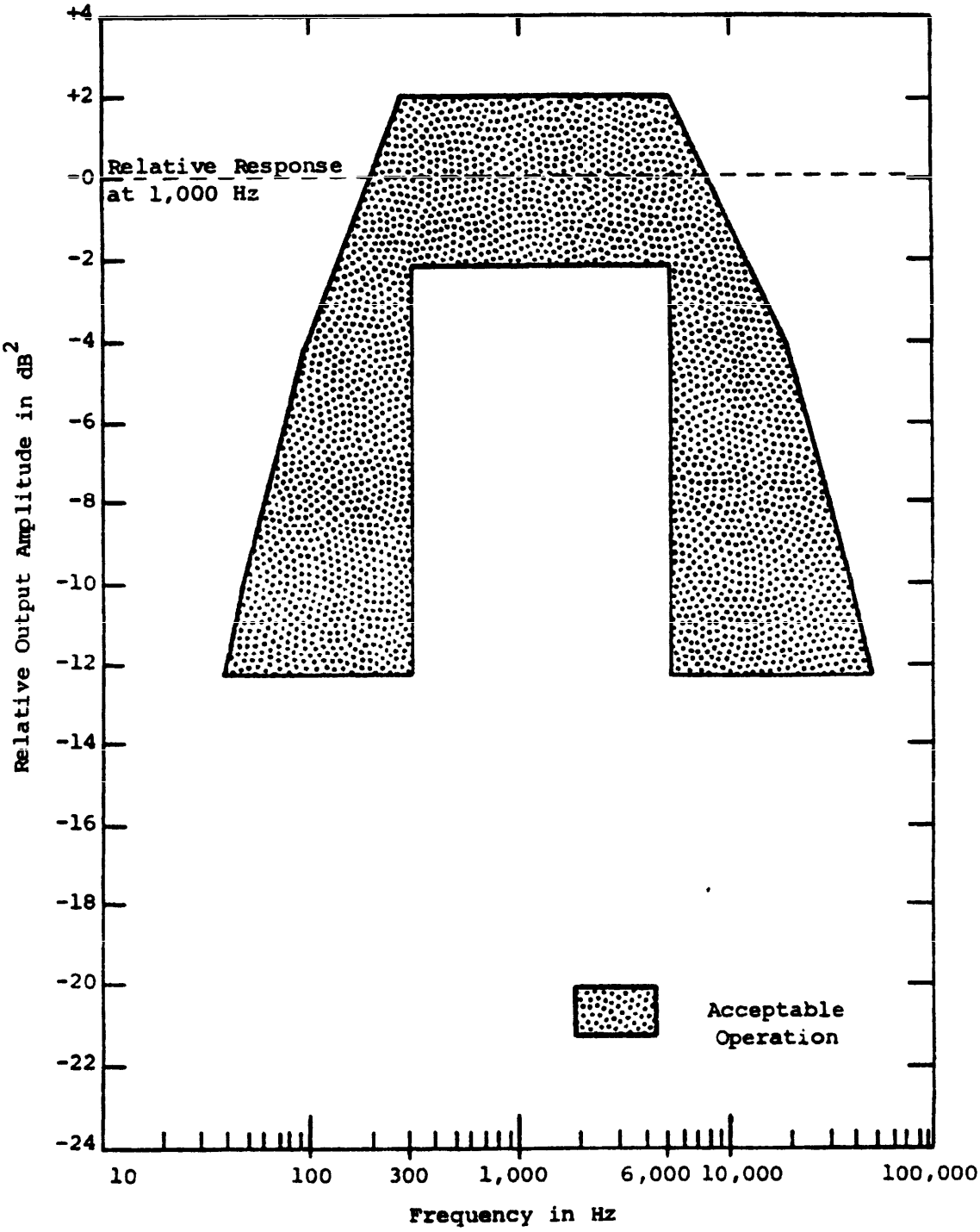


FIGURE 1. C-2646 functional diagram.

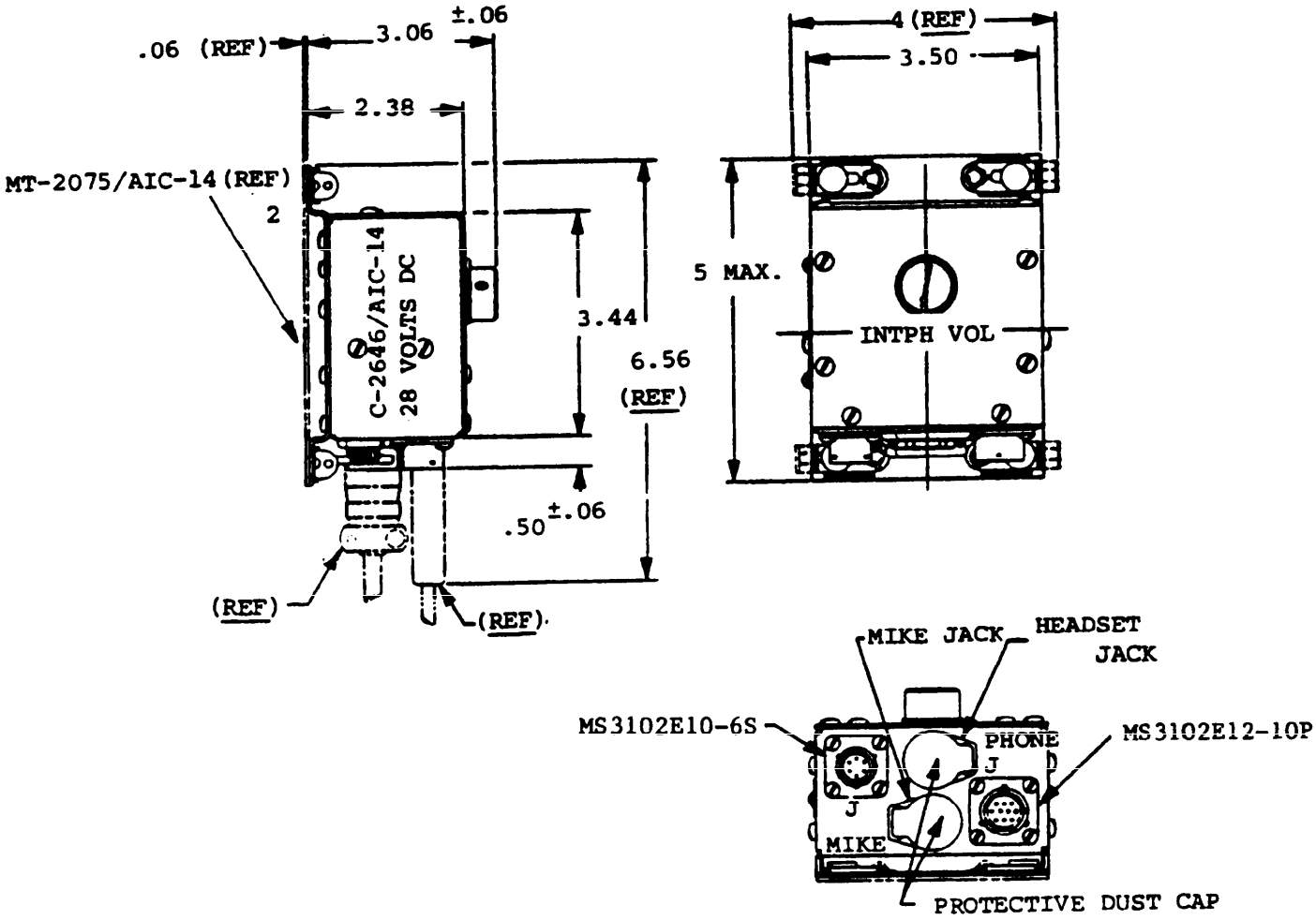
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1/ Termination = 600 ohms for AUDIO output.
2/ Output amplitude is measured with respect to the amplitude at 1,000 Hz.

FIGURE 2. Output response curve.

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

2/ Interphone control C-2646/AIC-14 is designed to be mounted on mounting plate MT-2075/AIC-14.

FIGURE 3. C-2646 outline drawing.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-22352/19(AS)	INTERCOMMUNICATIONS SETS, AN/AIC-14 AND AN/AIC-14A ACTIVE INTERPHONE CONTROL PANEL C-2646/AIC-14

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
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a. Paragraph Number and Wording:

b. Recommended Wording:

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